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Dendritic Growth into Undercooled Alloy Melts

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SUMMARY

Free dendrites growing in a liquid alloy reject both heat and (for $k_0 < 1$) solute. A growth equation has been developed both by coupling their diffusion fields for a parabolic tip and by applying a stability criterion. This relationship indicates that (1) the dendrite tip radius passes through a minimum with increasing solute concentration and (2) at a given undercooling the growth velocity should increase with increasing solute concentration and then decrease as higher solute levels are reached. These findings are shown to correspond to experimental results obtained using succinonitrile-acetone alloys.

1. INTRODUCTION

The cast structure of most alloys consists of both columnar and/or equiaxed dendrites. In order to improve the homogeneity of castings a certain proportion of equiaxed grains is advantageous. Therefore the enlargement of the equiaxed zone by inoculation or stirring of the melt has drawn much attention from researchers. Furthermore there is the possibility of growing columnar dendrites in an undercooled melt. Both types of growth are forms of free dendritic growth.

Free dendritic growth in pure substances, where heat transport is the controlling mechanism, has been reviewed by Huang and Glicksman [1]. Any model describing free dendritic growth in alloys must take into account the coupled processes of heat and mass transport *and* should also be applicable to the limiting case of the pure substance. Such models have been proposed by Temkin [2] and by Trivedi and Tiller [3] using the extremum approach.

During the last decade, however, it has become clear that the operating point of a dendrite may be determined by the limit of morphological stability of its tip rather than by extremum arguments. It was especially the work of Langer and Müller-Krumbhaar [4, 5] which showed that the dendrite tip is growing close to the condition of marginal stability. In view of this, Langer [6] developed a theory of alloy dendrite growth, but only for dilute solutions.

The aim of this paper is to present a new model which uses the stability criterion to predict the growth rate and the tip radius of a free dendritic crystal growing into an undercooled binary alloy melt.

2. THE MODEL

According to the stability criterion [4], the observed tip radius R is equal to the shortest wavelength λ_i which can form under the local growth conditions at the tip:

$$R = \lambda_i \tag{1}$$

Together with expressions describing the diffusion fields around the crystal tip, which depend on both the growth rate and the tip radius, eqn. (1) is used to find the growth rate and the tip radius under given growth conditions.

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Free dendritic crystals require an undercooled melt to sustain their growth because this acts as a sink for the latent heat that they reject during growth. When the distribution coefficient k_0 is less than unity, solute is also being rejected into the melt. The corresponding gradients of temperature and concentration in the liquid at the solid–liquid interface are therefore negative (Fig. 1).

Both heat and solute transport are functions of the total undercooling ΔT of the melt; as shown in Fig. 2, ΔT may be split into three parts, namely

$$\Delta T = \Delta T_t + \Delta T_c + \Delta T_R \tag{2}$$

The different terms on the right-hand side of eqn. (2) represent the thermal undercooling $\Delta T_t = T^* - T_\infty$, the solutal undercooling $\Delta T_c = T_l^R - T^*$ and the shift of the liquidus line in the phase diagram due to the finite radius of the tip which is given by $\Delta T_R = T_l^\infty - T_l^R$. Here T^* is the interface temperature at the tip, T_∞ is the bath temperature far from the tip, while T_l^∞ and T_l^R are the respective liquidus temperatures at a planar and at a curved interface.

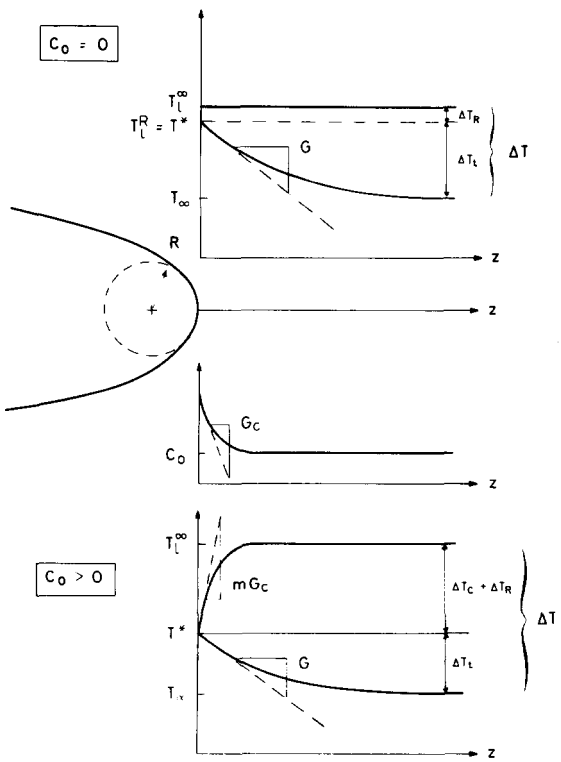

Fig. 1. Temperature and concentration fields ahead of a dendrite which is growing into an undercooled melt.

The general solution for the diffusion field around a shape-preserving paraboloid of revolution (which approximates the shape of a dendrite tip) with a steady state tip radius was found by Ivantsov [7] to be

$$\Omega = P \exp(P)\, E_1(P) \equiv \mathrm{Iv}(P) \tag{3}$$

where Ω is defined as the dimensionless supersaturation, $E_1(P)$ is the exponential integral function [8] and $\mathrm{Iv}(P)$ is the Ivantsov function. Equation (3) is the valid transport solution for both the solutal and the thermal fields, when the appropriate Peclet numbers are used.

For the thermal field,

$$P_t = \frac{VR}{2a} \tag{4}$$

where V is the interface growth rate, R is the tip radius and a is the thermal diffusivity. For the solutal field, the corresponding Peclet number is

$$P_c = \frac{VR}{2\tilde{D}} \tag{5}$$

where $\tilde{D}$ is the interdiffusion coefficient.

The dimensionless undercooling Ω_t corresponds to the thermal undercooling ΔT_t divided by a designated unit thermal undercooling $\Delta H/c_p$. Thus

$$\Omega_t = \frac{\Delta T_t\, c_p}{\Delta H} \tag{6}$$

where ΔH is the latent heat and c_p is the specific heat.

The scaling factor giving the dimensionless supersaturation Ω_c is the concentration differ-

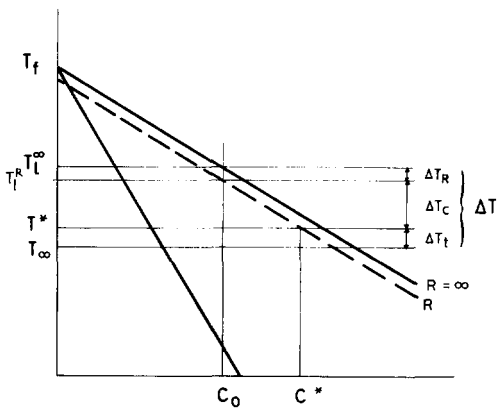

Fig. 2. Phase diagram and definition of related symbols.

ence $C^* - C_S^* = C^*(1-k_0)$ at the solid-liquid interface with C^* the concentration in the liquid at the tip, C_S^* the corresponding value in the solid and k_0 the equilibrium distribution coefficient. Therefore

$$\Omega_c = \frac{C^* - C_0}{C^*(1-k_0)} \tag{7}$$

where C_0 is the initial alloy composition.

Combining eqn. (6) with the Ivantsov solution (eqn. (3)) gives for the thermal undercooling

$$\Delta T_t = \frac{\Delta H}{c_p} \operatorname{Iv}(P_t) \tag{8}$$

The solutal (constitutional) undercooling ΔT_c is related to the concentration C^* at the solid-liquid interface via the phase diagram (Fig. 2):

$$\Delta T_c = m(C_0 - C^*) \tag{9}$$

where m is the liquidus slope, assumed to be constant. The interface concentration C^* can be expressed, using eqns. (7) and (3), as

$$C^* = \frac{C_0}{1-(1-k_0)\operatorname{Iv}(P_c)} \tag{10}$$

On insertion of eqn. (10) into eqn. (9), the expression for the solutal undercooling becomes

$$\Delta T_c = mC_0\left\{1 - \frac{1}{1-(1-k_0)\operatorname{Iv}(P_c)}\right\} \tag{11}$$

The depression of the liquidus temperature due to the curvature of a spherical interface of radius R representing the dendrite tip is given by

$$\Delta T_R = \frac{2\Gamma}{R} \tag{12}$$

where Γ is the Gibbs-Thomson parameter (the ratio of the solid-liquid interface energy to the melting entropy).

The total undercooling according to eqn. (2) can now be expressed using eqns. (8), (11) and (12) as

$$\Delta T = \frac{\Delta H}{c_p} \operatorname{Iv}(P_t) + mC_0\left\{1 - \frac{1}{1-(1-k_0)\operatorname{Iv}(P_c)}\right\} + \frac{2\Gamma}{R} \tag{13}$$

Equation (13) relates the product of the growth rate V and the tip radius R to the total undercooling ΔT of the melt.* Equation (13), however, is not sufficient to determine the unique values of V and R which define the operating state of the dendrite.

In order to obtain a second equation for V and R, the stability criterion (eqn. (1)) will be applied. According to Trivedi [9] and Kurz and Fisher [10] the stability criterion can be expressed as

$$R = \left\{\frac{\Gamma}{\sigma^*(mG_c - G)}\right\}^{1/2} \tag{14}$$

where G_c and G are the concentration gradient and temperature gradient respectively, evaluated at the solid-liquid interface. Here, σ^* is a stability constant which depends on the model used and is of the order of $1/4\pi^2 \approx 0.025$.

The gradients can be obtained by considering the heat and mass fluxes around the parabolic dendrite tip. The dimensionless concentration gradient in the liquid becomes $\tilde{G}_c = -2P_c$. The thermal gradient is expressed in the same way as $\tilde{G} = -2P_t$. Transforming both gradients into dimensional form gives

$$G_c = -\frac{2P_c C^*(1-k_0)}{R} \tag{15}$$

and

$$G = -\frac{2P_t\,\Delta H}{c_p R} \tag{16}$$

The conductivity-weighted average temperature gradient $\bar{G}$ is

$$\bar{G} = \frac{K_S G_S + K_L G_L}{K_S + K_L} \tag{17}$$

where K is the thermal conductivity and the subscripts S and L define solid and liquid respectively. For succinonitrile, $K_S \approx K_L$ and, for the isothermal dendrite represented by Ivantsov's solution, $G_S = 0$. Thus the mean temperature gradient becomes

$$\bar{G} = -\frac{P_t\,\Delta H}{c_p R} \tag{18}$$

*It should be noted that, if the radius R (eqn. (20)) is inserted into the $2\Gamma/R$ term, eqn. (13) can be solved for the product VR. This means that the Ivantsov solution already gives the tip temperature T^*.

Half the value given by eqn. (15) will be used for the concentration gradient G_c in eqn. (14) in order to obtain correspondence between the radius of eqn. (14) and the pure chemical case of Langer and Müller-Krumbhaar [5] (see Section 4). Therefore it is assumed that

$$\bar{G}_c = -\frac{P_c C^*(1-k_0)}{R} \tag{19}$$

C^* in eqn. (19) can be obtained from eqn. (10). By substituting the gradients from eqns. (18) and (19) into the stability criterion (14), a second relationship between V and R is obtained:

$$R = \frac{\Gamma/\sigma^*}{\dfrac{P_t\,\Delta H}{c_p} - \dfrac{P_c m C_0(1-k_0)}{1-(1-k_0)\,\mathrm{Iv}(P_c)}} \tag{20}$$

Equations (13) and (20) uniquely determine the growth rate and the radius of the dendrite tip for a fixed undercooling in a given alloy system. The determination of V and R requires the use of an iterative method such as the two-dimensional Newton scheme, where the differential quotients are approximated by difference expressions.

3. RESULTS

The tip radius R and the growth rate V were calculated using eqns. (13) and (20) for a given melt undercooling, over a range of concentrations of acetone in succinonitrile. These predictions are compared with measurements obtained by Chopra and Glicksman [11]. The thermophysical data of the succinonitrile-acetone system used in making the predictions are given in Table 1.

TABLE 1

Properties of succinonitrile-acetone mixtures [11]

T_f	331.24 K
c_p	1937.5 J kg^{-1} K^{-1}
ΔH	46.26 × 10^3 J kg^{-1}
a	1.14 × 10^{-7} m^2 s^{-1}
Γ	6.62 × 10^{-8} K m
D	1.27 × 10^{-9} m^2 s^{-1}
m	−2.16 (K (mol.%)$^{-1}$)
k_0	0.103 (mol.% (mol.%)$^{-1}$

Figure 3 gives predicted growth rate as a function of concentration for comparison with the measurements. This shows that small additions of solute to the pure material will *increase* the growth rate up to a maximum value. With further increase in concentration, the growth rate decreases continuously. The existence of the maximum is confirmed by the measurements [11, 12]. The present model thus correctly predicts the growth rate of succinonitrile-acetone dendrites over a considerable range of concentrations.

The increase in growth rate is related to a marked decrease in tip radius R, as shown in Fig. 4. After having reached a minimum value, the tip radius increases slowly with increasing concentration. At high concentrations the

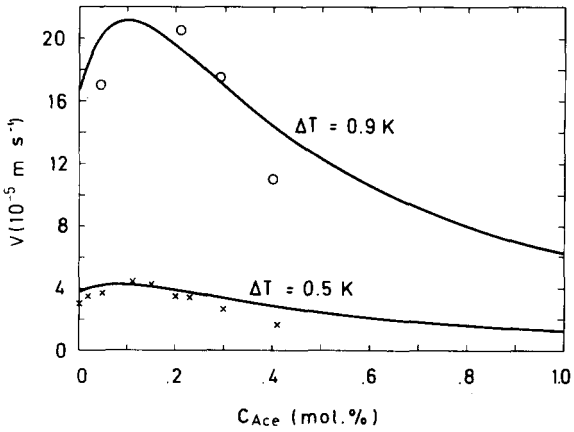

Fig. 3. Growth rate V for two undercoolings as a function of acetone concentration C_{Ace} ($\equiv C_0$) in succinonitrile-acetone alloys: ○, ×, measured; ——, calculated.

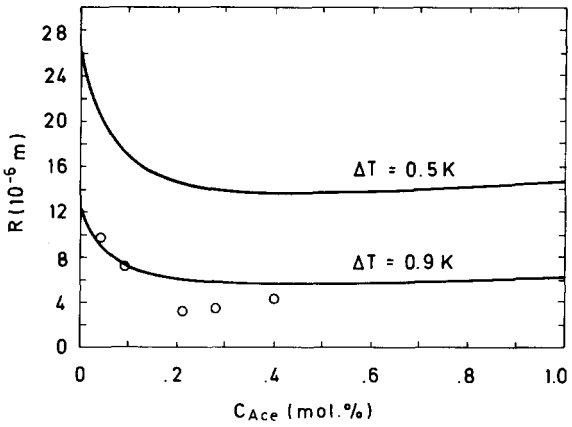

Fig. 4. Dendrite tip radius R for two undercoolings as a function of acetone concentration C_{Ace} ($\equiv C_0$) in succinonitrile-acetone alloys: ○, measured; ——, calculated.

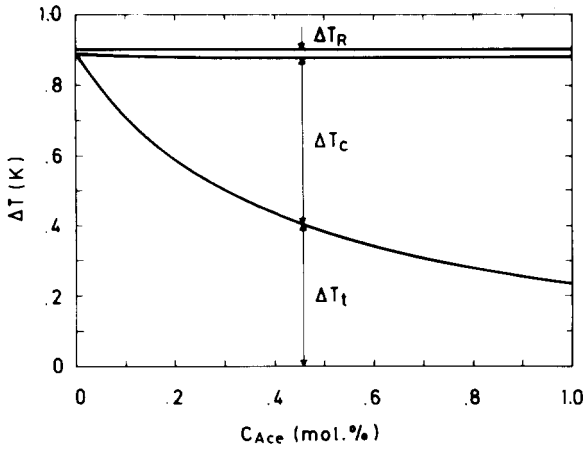

Fig. 5. Magnitude of the curvature undercooling ΔT_R, solutal undercooling ΔT_c and thermal undercooling ΔT_t for a total undercooling ΔT of 0.9 K as a function of acetone concentration C_{Ace} ($\equiv C_0$) in the succinonitrile-acetone system.

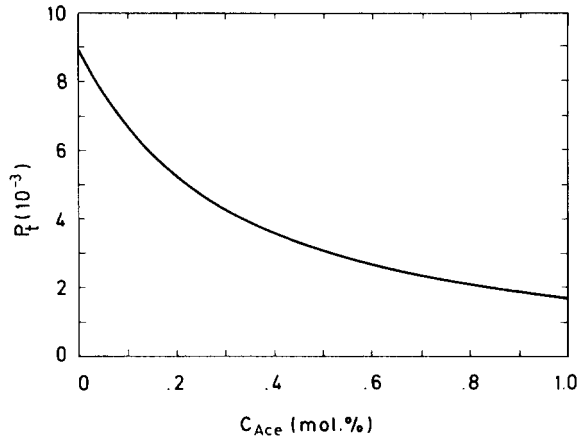

Fig. 6. Thermal Peclet number P_t as a function of acetone concentration C_{Ace} ($\equiv C_0$) in succinonitrile-acetone alloys ($\Delta T = 0.9$ K).

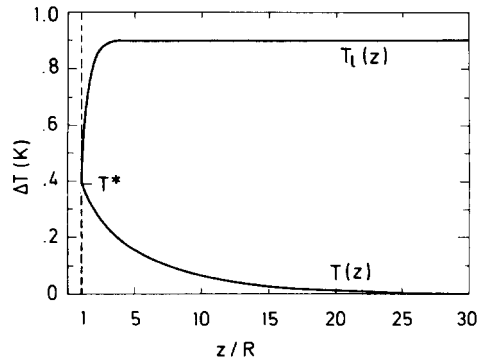

Fig. 7. Calculated temperature distributions ahead of a growing dendrite for a succinonitrile-(0.5 mol.% acetone) alloy with a total undercooling ΔT of 0.9 K.

model predicts R values which are larger than those observed experimentally. It is interesting to note that the concentration for maximum growth rate is lower than the concentration for minimum radius.

Neither minima nor maxima appear in the dependences on concentration of thermal or solutal undercoolings (Fig. 5) or Peclet number (Fig. 6), *i.e.* both ΔT_t and P_t decrease monotonically with increasing concentration. Finally, Fig. 7 shows the calculated temperature distributions ahead of the dendrite tip for a given alloy system as obtained by solving the equations for the diffusion fields [7].

4. DISCUSSION

Characteristic features of the model are the maximum in the growth rate and the minimum in the radius at certain concentrations. In order to understand this behavior, it is useful to consider the two gradients mG_c and G, and their difference which appears in the stability function (eqn. (14)). This has been done in Fig. 8. At very low solute concentrations, thermal processes predominate (*i.e.* the absolute value of the thermal gradient G is greater than the liquidus temperature gradient mG_c). At higher concentrations the liquidus temperature gradient is dominant. The maximum in the growth rate coincides with the

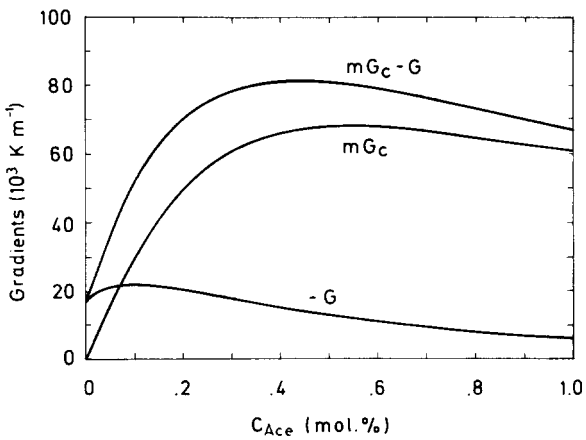

Fig. 8. Gradient terms of eqn. (14) as a function of acetone concentration C_{Ace} ($\equiv C_0$) in succinonitrile-acetone alloys ($\Delta T = 0.9$ K). It should be noted that the extremum in G corresponds to the maximum in the growth rate and the maximum of $mG_c - G$ defines the minimum in the radius.

maximum in $|G|$ in the regime controlled by heat flow (Fig. 5). The minimum in the radius is found at higher concentrations, where the denominator of eqn. (14), which is $mG_c - G$, has a maximum, *i.e.* where the driving force for the formation of instabilities is highest. A possible reason for the discrepancy between the calculated and observed R values of the alloys could be a change in one of the materials parameters such as the solid-liquid interface energy at small solute concentrations.

The undercooling of the interface (Fig. 5) and the Peclet number (Fig. 6) exhibit no extremum value because they do not depend (neglecting the ΔT_R term) on the stability criterion.

The behavior of the model will now be compared with the limiting cases described by Langer and Müller-Krumbhaar [5]. The thermal case is found by setting the chemical term of eqn. (20) equal to zero ($C_0 = 0$), leading to the expression

$$VR^2 = \frac{2}{\sigma^*} \frac{\Gamma a c_p}{\Delta H} \tag{21}$$

which is identical with the expression of Langer and Müller-Krumbhaar [5]:

$$R = \left(\frac{\delta d_0}{\sigma^*}\right)^{1/2} \tag{22}$$

with the diffusion length δ identically equal to $2a/V$ and the capillarity length d_0 identically equal to $|\Gamma c_p/\Delta H|$.

Setting the thermal term of eqn. (20) equal to zero ($\Delta H = 0$) leads to the following expression for the purely solutal case:

$$VR^2 = \frac{2}{\sigma^*} \frac{\Gamma D}{mC^*(k_0 - 1)} \tag{23}$$

which corresponds to eqn. (22) when $\delta \equiv 2D/V$ and the capillarity length for dilute solutions is given by $d_0 = \Gamma/mC^*(k_0 - 1)$ [13].

It should be noted that in order to get this correspondence a factor of 0.5 for the concentration gradient had to be used in eqn. (14). The origin of this correction is not clear at the moment but might be related to the assumption of an isothermal dendrite tip.

In Fig. 9 the product VR^2 of the present model (eqn. (20)) has been plotted as a function of concentration together with the thermal and solutal limits given by eqn. (22). It can be seen that, already at moderate concentrations (greater than 0.5 mol.% acetone in this case), the solutal term in the stability criterion is dominant.

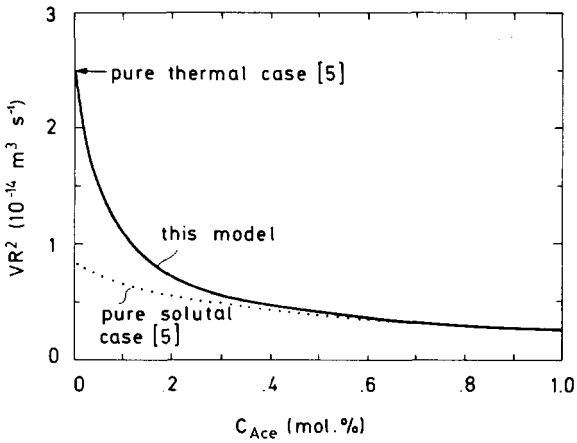

Fig. 9. Product of VR^2 as a function of acetone concentration C_{Ace} ($\equiv C_0$) in the succinonitrile-acetone system for an undercooling ΔT of 0.9 K. The limiting cases for purely thermal (pure substance) and for purely solutal dendrites are also indicated.

5. CONCLUSION

The thermal and solutal diffusion fields around the parabolic tip of an equiaxed or freely growing columnar dendrite have been coupled using Ivantsov's transport solution and a simplified stability criterion. In this way the complex behavior of freely growing alloy dendrites has been revealed. It is shown that the growth rate goes through a maximum at a certain alloy concentration while, at somewhat higher concentrations, the tip radius exhibits a minimum value. A similar behavior has been found in experiments using succinonitrile-acetone alloys. These findings which are not predicted by models using the extremum criterion provide another independent indication of the fact that stability criteria when coupled to appropriate transport solutions permit a good description of the observed behavior. The proposed model will be useful in predicting conditions for the columnar-to-equiaxed transition.

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